CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of first 8 positive odd numbers is: 1

(a) 56 **(b)** 72 **(c)** 64 **(d)** 128

2. The 10th term of the AP $7, 4, 1, -2, \ldots$ is: 1

(a) -23 **(b)** -17 **(c)** -30 **(d)** -20

3. The 12th term of $-5, -3, -1, 1, \ldots$ is: 1

(a) 17 **(b)** 19 **(c)** 15 **(d)** 24

4. The 6th term from the end of the AP with last term 150 and $d = 5$ is: 1

(a) 175 **(b)** 130 **(c)** 120 **(d)** 125

5. If $a_n = 3n + 7$, then the common difference is: 1

(a) 10 **(b)** 7 **(c)** 3 **(d)** 6

6. If $S_n = n^2 + 4n$, then $a_1 =$ 1

(a) 1 **(b)** 4 **(c)** 5 **(d)** 6

7. The sum of first 10 positive odd numbers is: 1

(a) 200 **(b)** 100 **(c)** 110 **(d)** 90

8. The 10th term of the AP $-4, 1, 6, 11, \ldots$ is: 1

(a) 41 **(b)** 46 **(c)** 36 **(d)** 50

9. The 9th term of $14, 17, 20, 23, \ldots$ is: 1

(a) 27 **(b)** 35 **(c)** 41 **(d)** 38

10. The 4th term from the end of the AP with last term 90 and $d = 6$ is: 1

(a) 78 **(b)** 66 **(c)** 72 **(d)** 108

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of the first 12 multiples of 5. 2

12. Find the sum of the AP $8, 12, \ldots, 64$. 2

13. Which term of the AP $-2, 2, 6, 10, \ldots$ is 34? 2

14. Find the sum of the first 15 multiples of 4. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, the 4th term is 16 and the 6th term is 28. Find the AP and its 9th term. 3

16. How many terms are there in the AP $9, 20, 31, \ldots, 174$? 3

17. Find the sum of the first 20 terms of the AP $8, \frac{17}{2}, 9, \frac{19}{2}, \ldots$ 3

OR

In an AP, the 5th term is 2 and the 9th term is -6. Find the AP and its 15th term. 3

18. How many terms are there in the AP $-7, -4, -1, \ldots, 80$? 3

OR

Find the sum of the first 14 terms of the AP $33, 30, 27, 24, \ldots$ 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How many terms of the AP $30, 27, 24, 21, \ldots$ must be taken so that the sum is 156? 5

OR

A loan is repaid in 10 monthly instalments; the first is ₹500 and each later instalment is ₹25 more than the previous. Find the instalment in the last month and the total amount repaid. 5

20. How many terms of the AP $5, 9, 13, 17, \ldots$ must be taken so that the sum is 324? 5

OR

A loan is repaid in 16 monthly instalments; the first is ₹50 and each later instalment is ₹5 more than the previous. Find the instalment in the last month and the total amount repaid. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND